Final Examination Practice Sheet - ANSWER KEY

Course Code: CSE252

Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Section(s): K3P25RB

Syllabus Covered: Unit 1 to Unit 6

Pattern: per unit, 8 MCQs (1 mark) and 2 long questions (10 marks)

Purpose: self-practice for the end-term examination

For the instructor. Every question is a reworded version of a question in the course question bank (scenario, names or numbers changed; numerical answers recomputed). The "Bank" column gives the source question number in that unit.

Unit 1: Foundations of Artificial Intelligence

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(B)**	The Turing Test. Proposed by Alan Turing in 1950.	Q3
2	**(D)**	Narrow AI. Narrow AI is bounded to one specialised task.	Q4
3	**(A)**	The gripper and motors (actuators). Actuators act; sensors perceive.	Q6
4	**(C)**	Fully observable. The sensors give the complete state.	Q7
5	**(C)**	Bias inherited from the historical training data. The model reproduces past unfair decisions.	Q16
6	**(A)**	Episodic. Independent decisions make the task episodic.	Q19
7	**(D)**	The incoming data has drifted, so the model needs monitoring and retraining. Data drift is caught in the monitoring stage.	Q22
8	**(B)**	Deep learning is a subset of machine learning that uses multi-layer neural networks. AI contains ML, which contains DL.	Q11

Part B - Expected answers and marking scheme

9. Explain the PEAS framework, and apply it to an autonomous lawn-mowing robot working on a university campus. *(10 Marks) [Bank Q1]*

(a) Explain what each of the four PEAS elements - performance measure, environment, actuators and sensors - is meant to capture.

Performance measure: the criteria by which success is judged. Environment: the surroundings the agent works in. Actuators: how the agent acts on the environment. Sensors: how it perceives the environment. (4 marks.)

(b) Write out what each of the four would contain for this robot.

P: area mowed, even grass height, time and battery used, no collisions with people, pets or benches, safety. E: lawns, paths, trees, flower beds, students, weather, charging dock. A: wheels and steering, cutting blade, blade-height motor, warning light or buzzer. S: GPS, cameras, LiDAR or ultrasonic sensors, bump sensors, grass-height sensor, battery sensor, rain sensor. (4 marks.)

(c) State whether the campus lawn is fully or partially observable, with one reason.

Partially observable: the sensors have limited range and cannot see behind trees or buildings, and people can appear suddenly, so the robot never knows the complete state of the lawn. (2 marks.)

10. Distinguish between Artificial Intelligence, Machine Learning and Deep Learning, using healthcare as the setting. *(10 Marks) [Bank Q2]*

(a) Describe the scope of each of the three and state how they are related to one another.

AI: the broad field of making machines perform tasks that need intelligence (reasoning, planning, perception), including rule-based systems. ML: a subset of AI in which systems learn patterns from data rather than following hand-written rules. DL: a subset of ML using multi-layer neural networks that learn their own features from raw data such as images. Relation: DL is inside ML, which is inside AI. (6 marks.)

(b) Give one healthcare application of each, and state in one sentence why it belongs where you placed it.

AI: a rule-based expert system for drug-interaction checking (explicit rules, no learning). ML: predicting hospital readmission from tabular patient records with a Random Forest (learned from data, hand-chosen features). DL: a convolutional network detecting tumours in MRI scans (learns features from raw pixels). (4 marks.)

Unit 2: Data Preparation and Machine Learning Workflow

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(A)**	0.30. (50 - 20) / (120 - 20) = 0.30.	Q13
2	**(C)**	2.5. (82.5 - 70) / 5 = 2.5.	Q14
3	**(B)**	Yes, as the upper fence is 80. IQR = 20; upper fence = 50 + 30 = 80, and 85 > 80.	Q15
4	**(D)**	750. Three of the four folds: 750 rows.	Q16
5	**(B)**	(4, 2). Summing over axis 1 removes the height dimension.	Q8
6	**(D)**	Information from the test set leaks into training, so the test score is optimistic. Fit imputers and scalers on the training set only.	Q12
7	**(A)**	One-hot encoding, since the categories have no order. Ordinal codes would invent a false order.	Q7
8	**(C)**	info(). info() summarises the DataFrame structure.	Q3

Part B - Expected answers and marking scheme

9. Apply a preprocessing pipeline to a hospital table that holds a blood-pressure column in which 5 per cent of the entries are missing and a few readings are obvious typing errors (such as 1,200), a categorical blood-group column with eight values, and an annual-income column ranging from 0 to 50 lakh rupees. *(10 Marks) [Bank Q1]*

(a) State the treatment you would apply to each of the three columns, with a one-line reason for each.

Blood pressure: detect impossible values with domain limits or the IQR rule and set them to missing, then impute with the median (robust to the extreme values). Blood group: one-hot encode, because the eight groups are nominal with no order. Income: scale (standardise or min-max; a log transform helps the skew) so its huge range does not dominate distance- or gradient-based models. (6 marks.)

(b) State whether scaling should be carried out before or after the train-test split, and give the reason.

After the split. Fit the imputer and scaler on the training set only and apply the fitted values to the test set. Fitting on all the data leaks test-set statistics into training and inflates the reported score. (4 marks.)

10. Examine and correct the following setup: an analyst splits a dataset 70 to 30, tunes a model by checking it repeatedly on the 30 per cent part until accuracy stops improving, and reports that accuracy as the expected performance on new data. *(10 Marks) [Bank Q2]*

(a) Identify the methodological error and explain what it does to the reported figure.

The test set has been used for tuning, so it is no longer unseen: the hyperparameters have been fitted to it. The reported accuracy is optimistically biased and will usually drop on genuinely new data. (5 marks.)

(b) Describe the corrected workflow, using cross validation on the training partition.

Split once into training and test sets and lock the test set away. Tune the hyperparameters with k-fold cross validation on the training set only (e.g. grid search with 5 folds). Refit the best configuration on the whole training set. Evaluate once on the untouched test set and report that figure. (5 marks.)

Unit 3: Supervised Machine Learning

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(D)**	15.2. 3.2 + 1.5 x 8 = 15.2.	Q13
2	**(B)**	7. 17 = 2 x 5 + c, so c = 7.	Q14
3	**(C)**	Class 1, since sigmoid(0.8) is about 0.69, above 0.5. A positive score always gives p > 0.5.	Q15
4	**(A)**	Pass. Three of five neighbours are Pass.	Q16
5	**(A)**	0.75. 0.09 / (0.03 + 0.09) = 0.75.	Q20
6	**(C)**	Overfitting; prune or limit its depth. A large train-test gap means high variance.	Q9
7	**(B)**	It assumes the features are independent of one another given the class. Conditional independence is rarely true.	Q6
8	**(D)**	KNN uses distances, so wide-range features dominate; a tree splits on one feature at a time by threshold. Scaling does not change the order of values within a feature.	Q10

Part B - Expected answers and marking scheme

9. Apply the bias-variance framework to three models trained on the same dataset. Model X has training error 30 per cent and test error 31 per cent. Model Y has training error 2 per cent and test error 20 per cent. Model Z has training error 8 per cent and test error 10 per cent. *(10 Marks) [Bank Q1]*

(a) For each model, state whether it is underfitting, overfitting or reasonably fitted, and justify your answer from its pair of error figures.

X: underfitting (high bias) - both errors high and close. Y: overfitting (high variance) - near-zero training error but a large gap to test error. Z: reasonably fitted - both errors low with a small gap. (6 marks.)

(b) State one change you would make to Model Y and the effect you expect it to have.

Any one: reduce complexity (limit depth, prune, fewer features), add L1/L2 regularisation, collect more training data, or use bagging/Random Forest. Expected effect: training error rises slightly, test error falls and the gap narrows, because variance is reduced. (4 marks.)

10. Compare logistic regression, K-Nearest Neighbour and Random Forest as classifiers for a drone that must decide, from ten sensor readings, whether a landing spot is safe. *(10 Marks) [Bank Q2]*

(a) For each of the three, state how it forms its decision boundary.

Logistic regression: weighted sum of the readings passed through a sigmoid; boundary is a hyperplane (linear) where p = 0.5. KNN: majority vote of the K nearest stored examples; boundary is non-linear and local, depends on K and scaling. Random Forest: many decision trees splitting on feature thresholds, combined by vote; boundary is made of axis-parallel regions and can be highly non-linear. (6 marks.)

(b) Recommend one of the three for use on board the drone, where a decision is needed within a few milliseconds, and justify the choice.

Logistic regression (or a small Random Forest): one weighted sum and a sigmoid over ten inputs takes microseconds and little memory. KNN must compute distances to every stored example at run time, so it is too slow and memory-hungry. A small, shallow forest is acceptable if the boundary is clearly non-linear. (4 marks.)

Unit 4: Unsupervised Learning

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(B)**	0.23. 115 / 500 = 0.23.	Q16
2	**(A)**	0.80. 200 / 250 = 0.80.	Q17
3	**(D)**	2. 45 + 30 = 75 per cent.	Q18
4	**(C)**	3 and 21. Groups {2, 4} and {20, 22}.	Q21
5	**(A)**	75 per cent. 60 / 80 = 75 per cent.	Q22
6	**(D)**	It is at most 0.20. Adding an item can never raise support.	Q15
7	**(C)**	Agglomerative merges bottom-up; divisive splits top-down. They build the hierarchy in opposite directions.	Q9
8	**(B)**	They lie in sparse regions and are left unassigned. Noise belongs to no cluster.	Q13

Part B - Expected answers and marking scheme

9. A car workshop logs the parts replaced at each service. Of 1,000 service visits, the air filter appears in 300, the oil filter appears in 250, and both appear together in 150. *(10 Marks) [Bank Q4]*

(a) Define support, confidence and lift.

Support: fraction of all transactions containing the itemset. Confidence of X -> Y: support(X and Y) / support(X). Lift: confidence / support(Y), i.e. how much more often X and Y occur together than if independent. (3 marks.)

(b) Compute the support, the confidence and the lift for the rule "if the air filter is replaced then the oil filter is replaced", showing the working.

Support = 150 / 1000 = 0.15. Confidence = 150 / 300 = 0.50. Support(oil filter) = 250 / 1000 = 0.25. Lift = 0.50 / 0.25 = 2.0. (5 marks.)

(c) Interpret the value of the lift you obtain.

Lift 2.0 > 1: an air-filter visit makes an oil-filter change twice as likely as on an average visit - a strong positive association, useful for stocking or bundled service offers. (2 marks.)

10. Compare k-Means, agglomerative hierarchical clustering and DBSCAN. *(10 Marks) [Bank Q2]*

(a) For each of the three, state the cluster shape it assumes and whether the number of clusters must be supplied in advance.

k-Means: convex, roughly spherical, similar-sized clusters; k must be given. Agglomerative: shape depends on linkage (single allows elongated, complete gives compact); k is not needed in advance - the dendrogram is cut afterwards. DBSCAN: arbitrary shapes following dense regions; k is not needed, but eps and MinPts are. (6 marks.)

(b) A survey drone records LiDAR points that trace curved river banks, together with some isolated spurious readings. Recommend the most suitable of the three and justify the choice.

DBSCAN: it follows the connected dense points along curved banks of any shape and labels the isolated spurious readings as noise. k-Means would cut the curves into round blobs and force the spurious points into clusters; hierarchical clustering is costly on large point clouds and single linkage would chain through the noise. (4 marks.)

Unit 5: Model Evaluation and Improvement

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(C)**	0.60. 30 / (30 + 20) = 0.60.	Q15
2	**(A)**	0.62. 2 x 0.8 x 0.5 / 1.3 = 0.615.	Q17
3	**(B)**	300. 1,500 / 5 = 300.	Q24
4	**(D)**	Predicting the majority class alone already scores 99 per cent. Use precision, recall or F1 instead.	Q8
5	**(D)**	It rises, because more cases are called positive. Fewer positives are missed.	Q11
6	**(B)**	1.0. Perfect ranking of positives above negatives gives AUC = 1.	Q12
7	**(A)**	Precision is measured over the predicted positives; recall over the actual positives. TP/(TP+FP) versus TP/(TP+FN).	Q9
8	**(C)**	The data has drifted over time. New data no longer matches the training distribution.	Q23

Part B - Expected answers and marking scheme

9. A model that screens chest X-rays for a disease is evaluated on 2,000 X-rays, of which 100 are genuinely positive. The model flags 150 X-rays as positive, and 80 of those flagged are genuinely positive. *(10 Marks) [Bank Q1]*

(a) Construct the confusion matrix.

TP = 80, FP = 150 - 80 = 70, FN = 100 - 80 = 20, TN = 2000 - 80 - 70 - 20 = 1830. (3 marks.)

(b) Compute the accuracy, the precision and the recall, showing the working.

Accuracy = (80 + 1830) / 2000 = 0.955. Precision = 80 / 150 = 0.533. Recall = 80 / 100 = 0.80. (5 marks.)

(c) State which single metric matters most here, given that a missed case is far more serious than an extra review.

Recall: it measures how many actual positive cases are caught; missed cases (false negatives) are the costly error, so the threshold should be set to keep recall high. (2 marks.)

10. Compare grid search and random search as strategies for hyperparameter tuning. *(10 Marks) [Bank Q2]*

(a) State how each of the two explores the search space.

Grid search tries every combination of the listed values exhaustively; its cost multiplies with each added parameter. Random search samples a fixed number of combinations at random from given ranges or distributions; it covers more distinct values of the important parameters for the same budget. (5 marks.)

(b) Compute the total time needed to search an exhaustive grid of 3 by 4 by 5 values for a gradient-boosting model, where a single fit takes 30 seconds and 4-fold cross validation is used. Show the working.

Combinations = 3 x 4 x 5 = 60. Fits = 60 x 4 folds = 240. Time = 240 x 30 s = 7,200 s = 120 minutes = 2 hours (plus one final refit). (5 marks.)

Unit 6: AI for Intelligent Systems and Robotics

Part A - MCQ answers

Q	Answer	Reason / working	Bank
1	**(A)**	7. 4 + 0.5 x 2 + 0.25 x 8 = 7.	Q16
2	**(D)**	0.50. Union = 100 + 80 - 60 = 120; 60 / 120 = 0.50.	Q17
3	**(C)**	5. drone, flies, high, lands, low.	Q18
4	**(B)**	3. 0.91, 0.66 and 0.72 pass.	Q21
5	**(B)**	No, the total of 130 ms exceeds 120 ms. 50 + 40 + 40 = 130 ms.	Q23
6	**(A)**	Image classification. Detection would also return a box.	Q5
7	**(D)**	The input distribution has shifted away from the training data. Night images were never seen in training.	Q13
8	**(C)**	It may stay stuck on a poor route because better routes are never tried. Pure exploitation can lock in a poor route; some exploration is needed.	Q10

Part B - Expected answers and marking scheme

9. Apply the reinforcement learning framework to a drone that must deliver medicines from the campus pharmacy to a hostel while avoiding trees and buildings. *(10 Marks) [Bank Q1]*

(a) Define the agent, the environment, the state, the set of actions and the reward function for this task.

Agent: the drone's controller. Environment: the campus airspace with trees, buildings, wind, the pharmacy and hostel. State: position, altitude, velocity, battery, distances to obstacles, package status. Actions: move forward/back/left/right, ascend/descend, hover, land. Reward: large positive for on-time delivery, small negative per time step or energy used, large negative for collision or crash, penalty for leaving the permitted area. (6 marks.)

(b) Explain the trade-off between exploration and exploitation as it applies to this drone.

Exploitation follows the best route found so far; exploration tries new routes or altitudes that may be faster. Too little exploration can lock in a poor route; too much wastes battery and risks collisions. Use epsilon-greedy with decaying epsilon, and explore in simulation rather than in the real campus. (4 marks.)

10. Compare image classification with object detection as the vision component of a library robot that must stop for people and trolleys in its aisles. *(10 Marks) [Bank Q2]*

(a) State what each of the two produces as its output and what each requires of its training data.

Classification: one label (with a probability) for the whole image; training data needs only an image-level label. Detection: a set of bounding boxes, each with a class and a confidence, locating every object; training data needs a box drawn around every object plus its class. The robot needs detection, because it must know where the obstacle is. (6 marks.)

(b) A predicted box of area 120 square pixels overlaps a ground-truth box of area 100 square pixels by 66 square pixels. Compute the intersection over union, showing the working, and state whether the detection is accepted at a threshold of 0.5.

Union = 120 + 100 - 66 = 154. IoU = 66 / 154 = 0.43. Since 0.43 < 0.5, the detection is not accepted (counted as a false positive). (4 marks.)

----- End of Answer Key -----